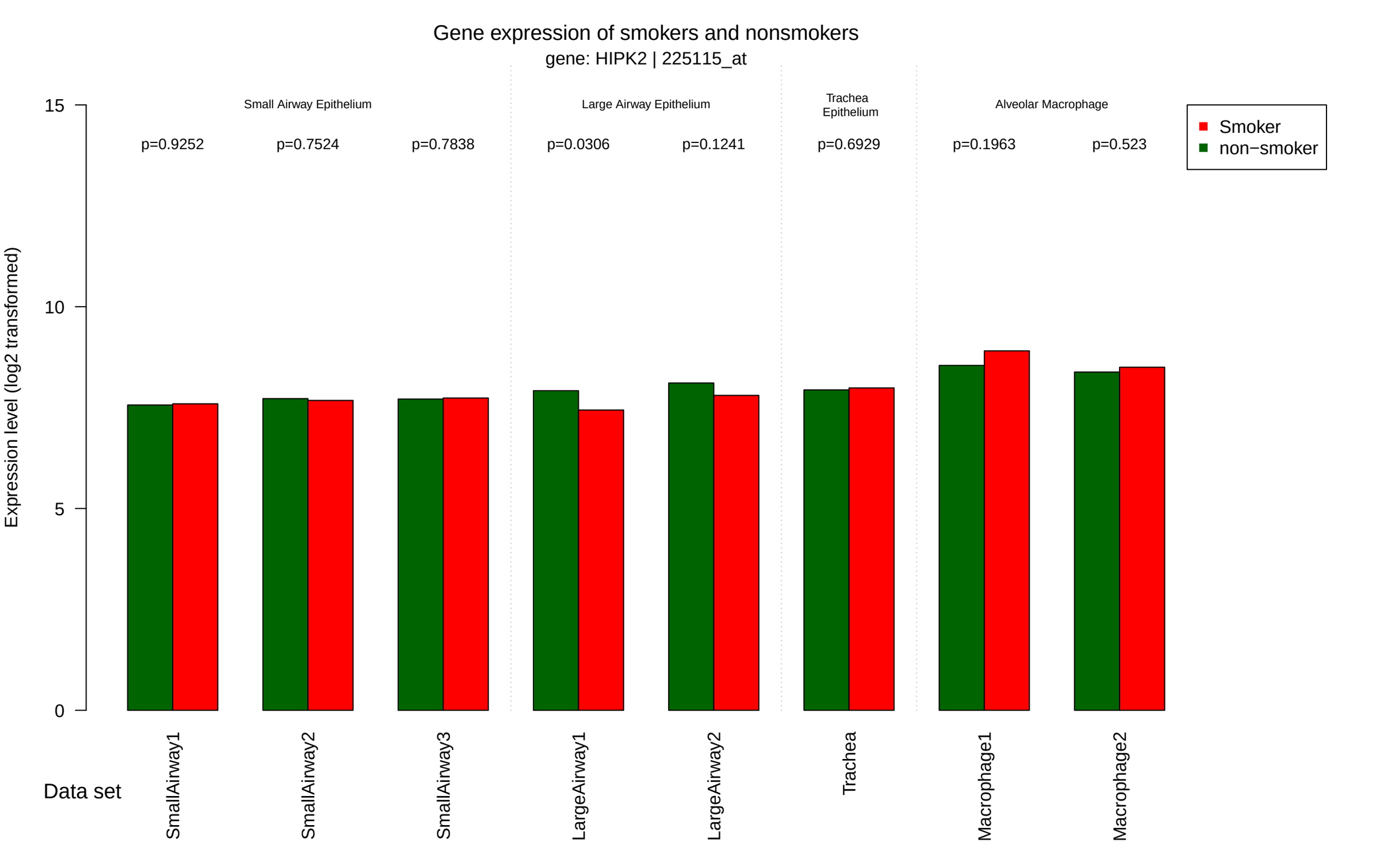
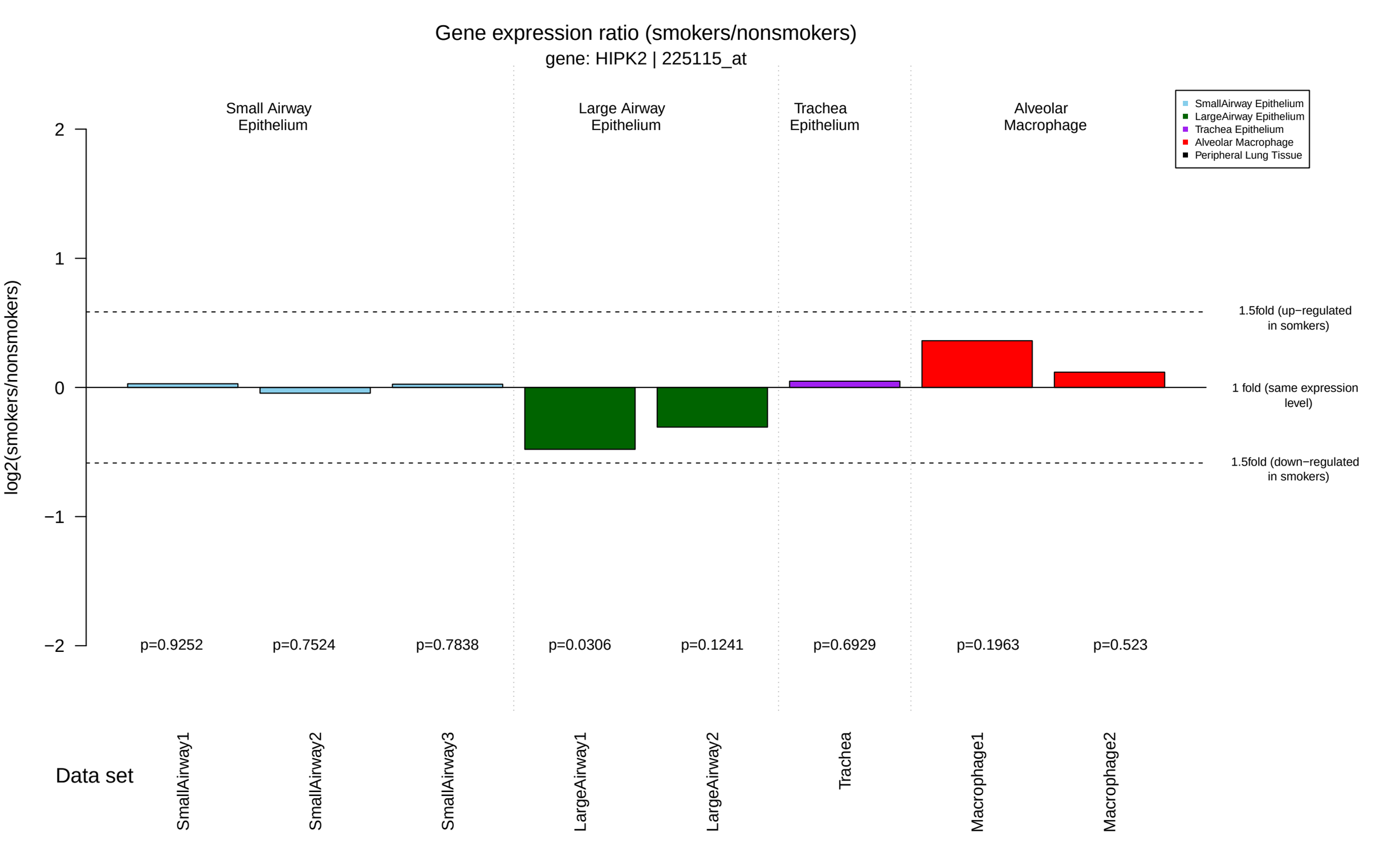
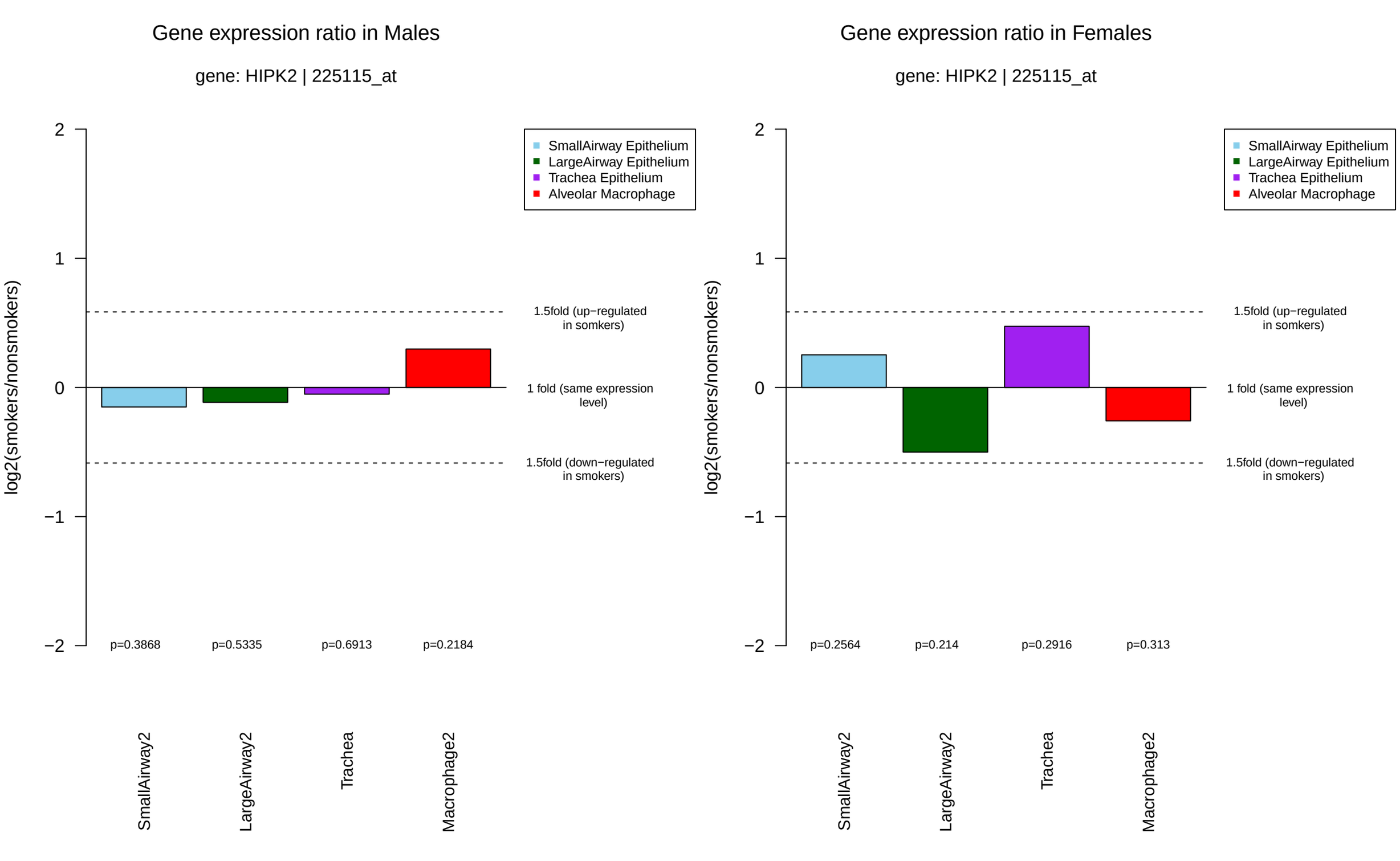
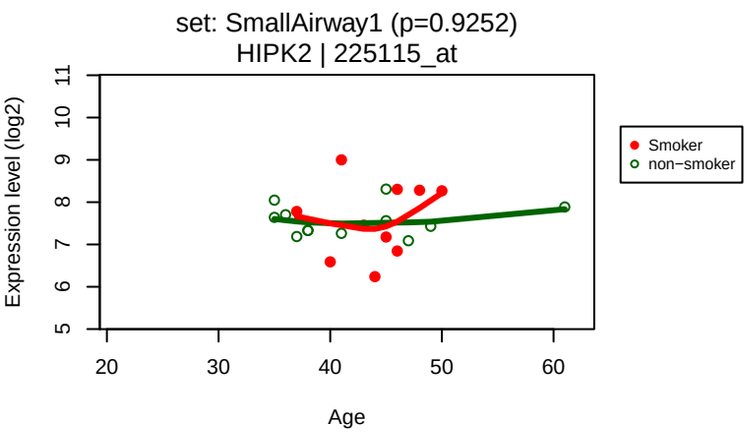
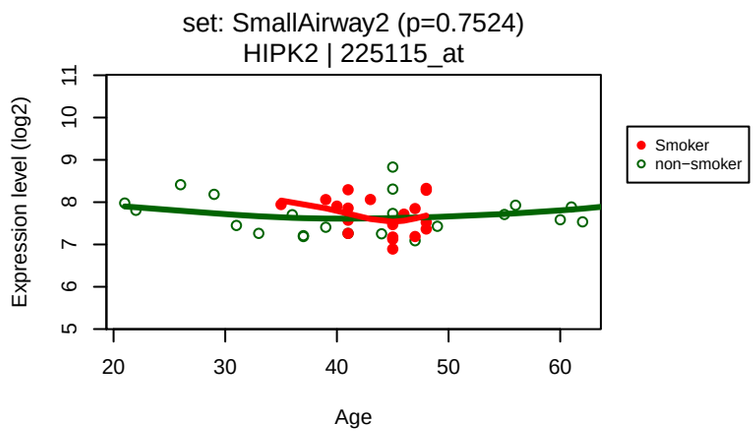
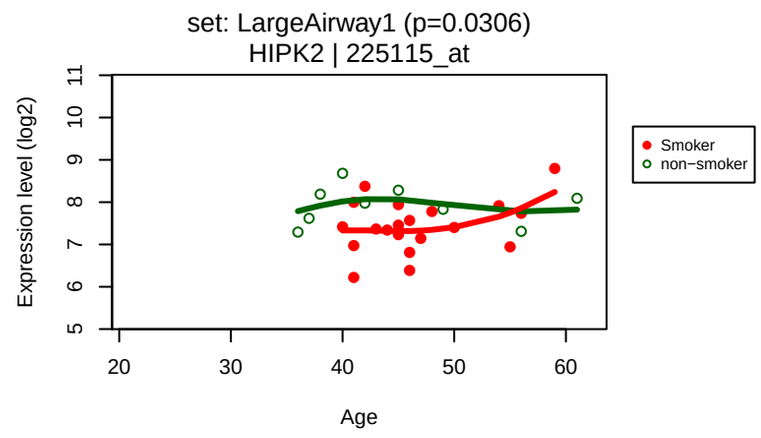

Gene expression of smokers and nonsmokers
gene: HIPK2 | 225115_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9252
p=0.7524
p=0.7838
p=0.0306
p=0.1241
p=0.6929
p=0.1963
p=0.523
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

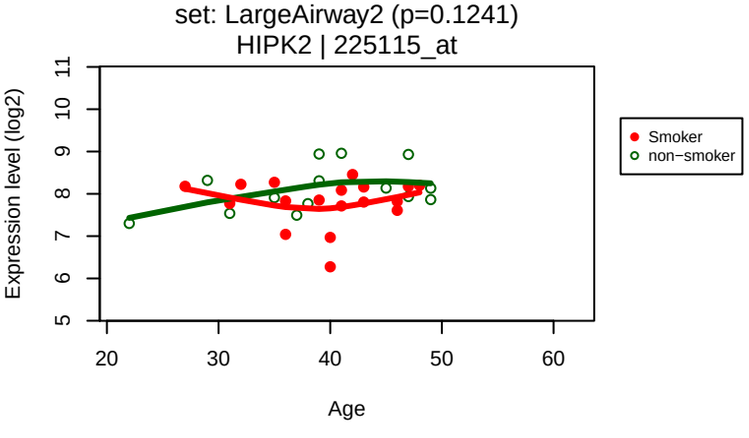

Gene expression ratio (smokers/nonsmokers)
gene: HIPK2 | 225115_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9252
p=0.7524
p=0.7838
p=0.0306
p=0.1241
p=0.6929
p=0.1963
p=0.523
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

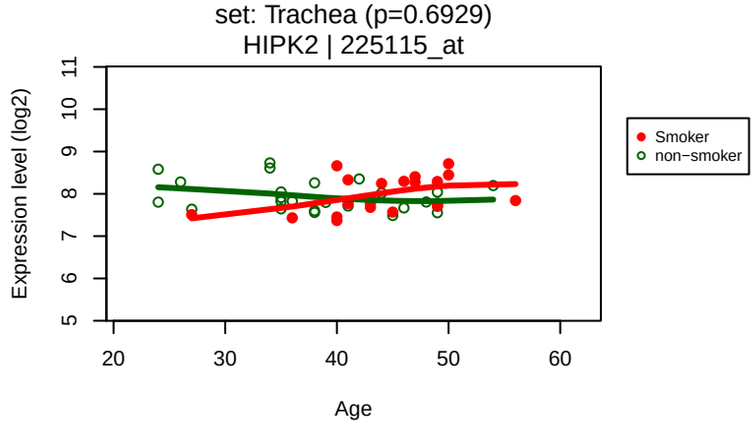

Gene expression ratio in Males
gene: HIPK2 | 225115_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3868
p=0.5335
p=0.6913
p=0.2184
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HIPK2 | 225115_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2564
p=0.214
p=0.2916
p=0.313
SmallAirway2
LargeAirway2
Trachea
Macrophage2

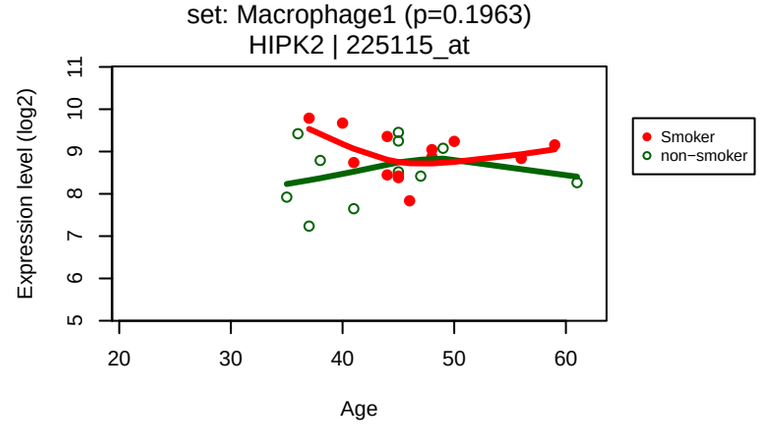

set: SmallAirway1 (p=0.9252)
HIPK2 | 225115_at
Expression level (log2)
Age
Smoker
non−smoker

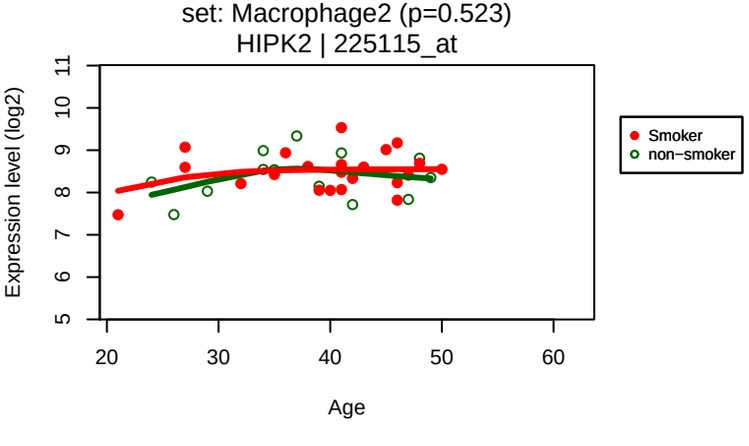

set: SmallAirway2 (p=0.7524)
HIPK2 | 225115_at
Expression level (log2)
Age
Smoker
non−smoker

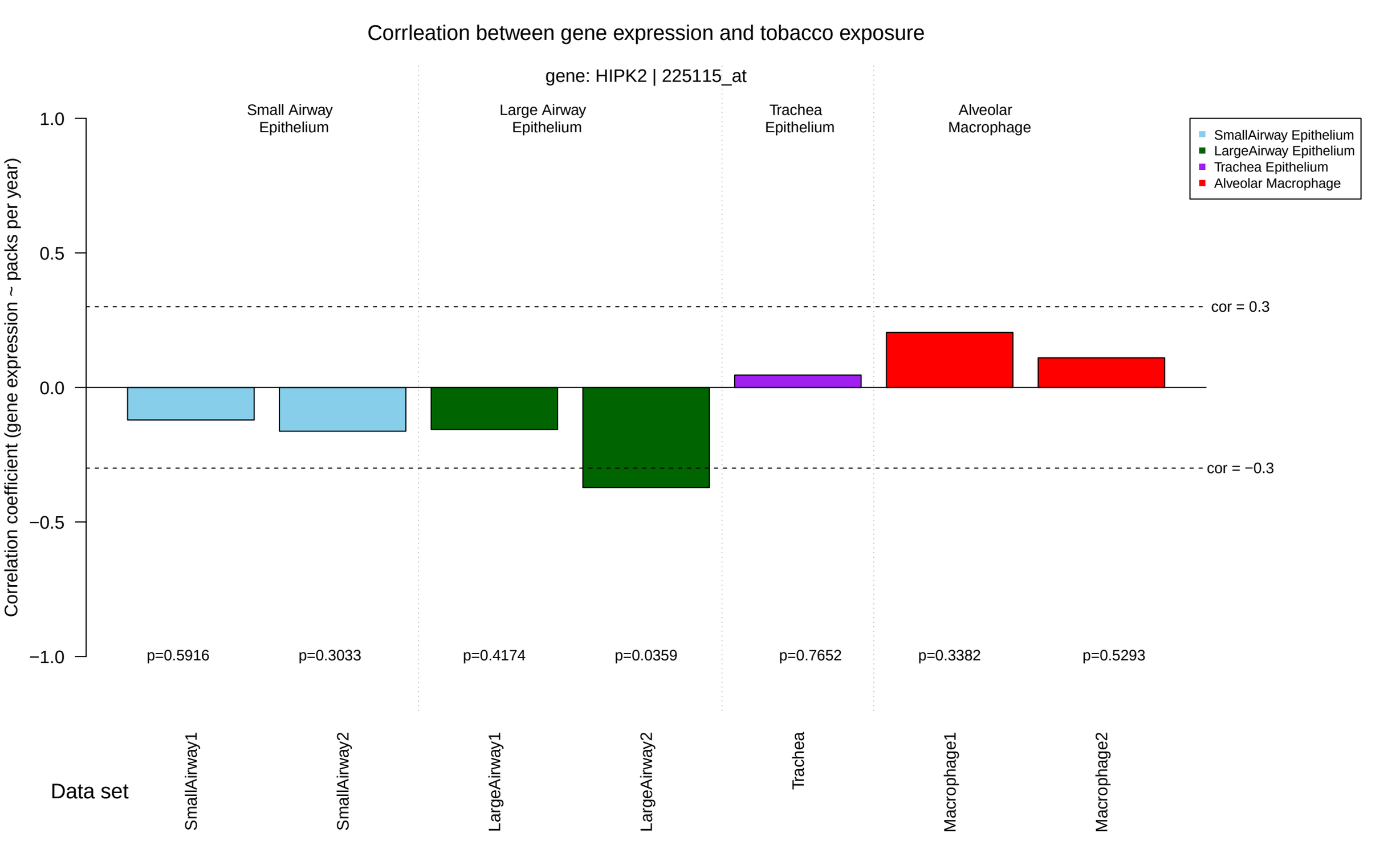

set: LargeAirway1 (p=0.0306)
HIPK2 | 225115_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1241)
HIPK2 | 225115_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6929)
HIPK2 | 225115_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1963)
HIPK2 | 225115_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.523)
HIPK2 | 225115_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HIPK2 | 225115_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5916
p=0.3033
p=0.4174
p=0.0359
p=0.7652
p=0.3382
p=0.5293
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2